Bonded anchor pryout test model

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1 Introduction

Regarding the pryout tests for bonded anchors the model in Figure 1 is given for the process of virtual approval. In the Figures and throughout the text quantities with indices $(\cdot)_\mathrm{c}, (\cdot)_\mathrm{s}, (\cdot)_\mathrm{m}, (\cdot)_\mathrm{a}$ indicate geometric/ material properties of the concrete, steel, mortar and anchor respectively.

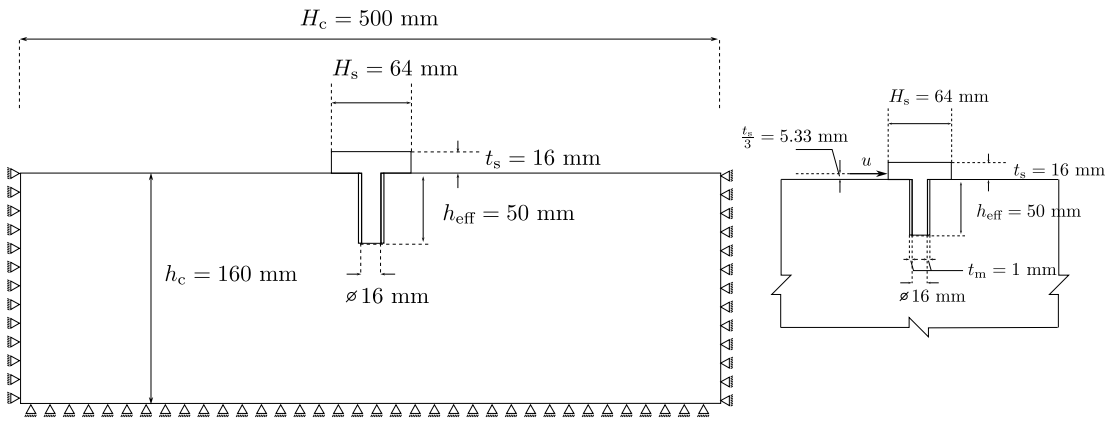

Figure 1: Geometry of the bonded anchor model. Left: Full model geometry and boundary conditions for concrete slab. Right: Detail near the anchor, including a) boundary conditions: horizontal displacement u on the steel plate with eccentricity, b) mortar thickness.

1.1 Geometry

The model consists of a cylindrical steel anchor with diameter $d_\mathrm{a} = 16$ mm, embedded till depth $h_\mathrm{eff} = 50$ mm, in an orthogonal parallelepiped concrete slab with dimensions (length: $H_\mathrm{c} = 500$ mm, width: $H_\mathrm{c} = 500$ mm, height: h_c= 160 mm). A cylindrical hole is present inside the concrete slub with diameter $d_\mathrm{h} = d_\mathrm{a} + 2t_\mathrm{m} = 18$ mm and $t_\mathrm{m} = 1$ mm is the mortar thickness. No modeling of the drilling process or any of the residual stresses that may develop takes place. The anchor is placed inside the hole so that their axes along the height match. The mortar region is then modeled as a hollow cylinder sleeve with height h_eff and constant thickness t_m between the anchor and the surrounding concrete.

The anchor is connected to a steel plate on top with dimensions (length: $H_\mathrm{s} = 4d_\mathrm{a} = 64$ mm, width: $H_\mathrm{s} = 4d_\mathrm{a} = 64$ mm, height: $t_\mathrm{s} = d_\mathrm{a} = 16$ mm). The plate contains a hole in the middle with diameter matching the diameter of the anchor, through its thickness. The connection of the anchor to the plate is as follows: First, the anchor is extended above the concrete slab by a length equal to the thickness of the steel plate. Second, the anchor extension is placed in the center of the steel plate hole. The two components (anchor, plate) then are merged together behaving as one body for subsequent discretizations of the model.

The assumed loading and the model are symmetric regarding the global z coordinate thus only half of the model needs to be simulated (see also Figure 2).

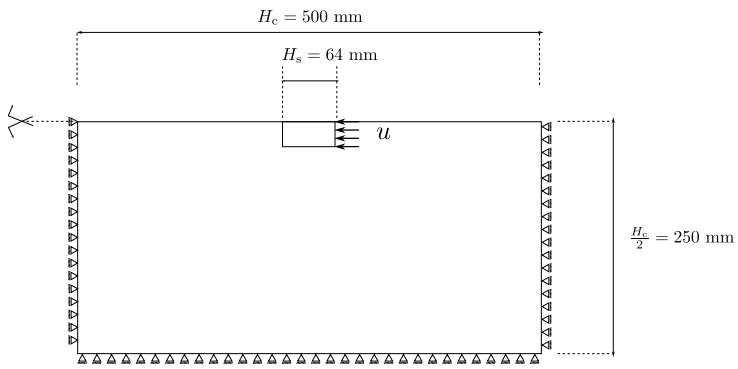

Figure 2: Top view of the model including the plane of symmetry across which the model and the loading are symmetric.

the component dimensions are summarized in the next table:

Parameter	**Value**	**Description**
d_a	16 mm	Anchor diameter
h_{eff}	50 mm	Effective embedment depth
t_m	1 mm	Mortar thickness
H_c	500 mm	Slab length and width
h_c	160 mm	Slab height
H_s	64 mm	Plate length and width
t_s	16 mm	Plate thickness

Table 1: Geometrical parameters of the bonded anchor model.

1.2 Material properties

The concrete properties are given in Table 2, i.e grade C20/25. In the Table: E_c denotes the Young's modulus, ν_c denotes the Poisson's ratio, f_{cy} denotes the yield strength, f_{cu} denotes the ultimate strength, f_{bu} denotes the concrete ultimate biaxial strength, f_{tu} denotes the concrete ultimate tensile strength, G^f_{Ic} is the concrete fracture energy in mode one fracture, and ρ_c is the density of concrete.

The anchor and steel plate are made from structural steel with known elastic properties and density see Table 3. In the Table: E_s denotes the Young's modulus, ν_s denotes the Poisson's ratio and ρ_s is the density of steel.

Mortar properties are given in Table 3. For pryout tests failure in mortar is not critical therefore assume mortar to be elastic. Mortar stiffness differs in the normal and tangential directions. E_{nm}, G_{nt}.

Parameter	**Value**
E_c	2.58×10^4 [MPa]
ν_c	0.18
f_{cy}	10.33 [MPa]
f_{cu}	31 [MPa]
f_{bu}	35.96 [MPa]
f_{tu}	2.7 [MPa]
G^f_{Ic}	136.63×10^{-3} [N/mm]
ρ_c	2.24×10^{-6} [kg/mm^3]

Table 2: Concrete properties

Parameter	**Value**
E_s	2×10^5 [MPa]
ν_s	0.33
ρ_s	7.85×10^{-6} [kg/mm^3]
E_{nm}	5400 [MPa]
G_{tm}	2700 [MPa]
ρ_m	2×10^{-6} [kg/mm^3]

Table 3: Steel and mortar properties

1.3 Boundary conditions

A uniform horizontal displacement boundary condition is applied on the left part of the steel plate at $t_s/3$ from the concrete surface. Other than that the plate is left free to rotate due to the eccentric displacement field applied.

The concrete slab is laterally constrained in all directions at its boundaries.

2 Example finite element discretization

The model is constructed and discretized with the help of Cubit [1] and then exported to Abaqus [2] format.

20 node quadratic brick finite elements with reduced integration for the concrete (C3D20R), linear 8 noded brick finite elements for the steel plate and anchor and linear 8 noded cohesive elements (COH3D8) for the mortar are used.

The connection between the steel plate and the concrete is modeled by a hard contact interface law, i.e interpenetration is not allowed and the friction between the two bodies is not taken into account. The lateral connection between 1) anchor and mortar and 2) mortar and concrete is performed using tie constraints. The base of the anchor at the embedment height is free.

Mortar is modeled with the use of traction-separation bond law available in Abaqus including damage see Table 4. Table contains, k_{nm} mortar stiffness ([MPa/mm]) in normal direction, k_{tm} mortar stiffness ([MPa/mm]) in the tangential direction (same for both tangential directions), and mortar density ρ_{m}.

Parameter	**Value**
k_{nm}	5400 [MPa/mm]
k_{tm}	2700 [MPa/mm]
ρ_{m}	2×10^{-6} [kg/mm^3]

Table 4: Mortar properties

2.1 Solution procedure

An implicit dynamic analysis is performed using the Backward Euler solver of Abaqus, in one step with adaptive time incrementation.

References

[1] Coreform LLC. Coreform cubit, 2025. Computer software.

[2] Dassault Systèmes SIMULIA. *Abaqus 2024 Documentation*. Dassault Systèmes, 2024. Finite element analysis software.